

Unit 2A Day 10 Notes

Today we will model periodic motion with trig functions 😊

Example 1: A Ferris wheel is 160 meters in diameter and boarded at its lowest point (6 O'Clock) from a platform which is 8 meters above ground. The wheel makes one full rotation every 16 minutes, and at time $t = 0$, when the wheel starts spinning, you are at the loading platform.

a. What is the period and what does it represent in this situation?

b. What is the equation of the midline?

c. What is the amplitude and what does it represent in this situation?

d. Write an equation to model the Ferris wheel's height above the ground as a function of time.

e. Use the information above to graph one period of the sinusoidal function.

Example 2: Over the past several years, the owner of a boutique on Aspen Avenue has observed a pattern in the amount of revenue for the store. The revenue reaches a maximum of about \$41,000 in April and a minimum of about \$26,000 in October. Suppose the months of the year are numbered 1 through 12.

a. What is the period and what does it represent in this situation?

b. What is the range and what does it represent in this situation?

c. Write a function that models the boutique's revenue during the year, where x corresponds to the month.

d. Use the information above to graph one period of the sinusoidal function.

You try! 😊 A buoy floating in the sea is bobbing in simple harmonic motion with period of 2 seconds and amplitude of 9 inches. Its displacement d from sea level at time $t=0$ seconds is -9 inches, and initially it moves upward (i.e. in the positive direction). The buoy bounces back and forth from a negative displacement to a positive displacement. Graph one period of the function. Then give the equation modeling the displacement d as a function of time t .